

Example Class MATH2101

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Introduction

This is the material used for MATH2101 (linear algebra I) in fall, 2024. The purpose of the problems is to clarify some concepts and possible strategies that are hopefully useful for your study. Though this example class is technically optional, there are some questions labeled (*Optional*) being more optional than others. They exist mainly because I personally reckon them helpful when understanding the concepts, and there seems to be a tradition of giving some insight for future courses in example classes. In case you find them too difficult, or you have a tight schedule, just skip them. They will not be the major focus in revision classes.

Most problems are not original and part of them come from the example sheet given by my predecessor Lai Ruize. I express my sincere gratitude to him here. Please also excuse any possible typos or mistakes.

Remarks on example class 4

The final example class will cover orthogonality and gives a list of problems to prepare you for the final. Further information is available at section 8.

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1 Matrix Algebra and Linear Equations

Problem 1.1. Let $A = \begin{pmatrix} 10 & 20 & 30 & 40 \\ 2 & 4 & 6 & 9 \\ 1 & 3 & 4 & 5 \\ 1 & 2 & 4 & 5 \end{pmatrix}$. Find the (2,1)-entry of the reduced row echelon form of A .

Problem 1.2. Multiple choices. More than one answer can be correct for the following questions.

1. Let A and B be matrices such that AB is well-defined. If AB has (), so does B .
A. a zero row
B. n rows
C. only integer entries
D. two identical columns.
2. Let A be an invertible $n \times n$ matrix. Which of the following must be true?
A. $A^T A$ is invertible
B. $Ax = e_1$ for some $x \in \mathbb{R}^n$
C. The set $\{Ae_1, Ae_2, \dots, Ae_n\}$ must be linearly independent
D. The rows of AB are linearly independent for any $n \times m$ matrix B .

Problem 1.3. Determine whether the following statements are true or false, and give simple explanations:

1. If $A^5 = O$, then so is A^4 .
2. Suppose the reduced row echelon form of A has a zero row, then the system $A\mathbf{x} = \mathbf{b}$ has infinitely many solutions if it's consistent.
3. If A^3 is invertible, so is A^4 . And is it still the same for A^4 and A^5 ?
4. If $\{\mathbf{u}, \mathbf{v}\}$ and $\{\mathbf{u}, \mathbf{w}\}$ are linearly dependent, then so is $\{\mathbf{v}, \mathbf{w}\}$.
5. Any square matrix of size larger than 1×1 can be written as the sum of two non-invertible matrices.
6. If a system of linear equations has more equations than unknown variables, then it cannot have infinitely many solutions.

Problem 1.4. Given the reduced row echelon form of $A = \begin{pmatrix} 2 & a & 7 \\ 6 & b & -6 \\ 8 & c & 1 \end{pmatrix}$ is $\begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & -3 \\ 0 & 0 & 0 \end{pmatrix}$, find the value of a , b , and c .

Problem 1.5. Compute the inverse of the following matrix without using row operations (i.e. using the adjugate).

$$\mathbf{M} = \begin{pmatrix} 3 & 4 & 5 \\ 1 & 0 & 2 \\ -2 & 4 & -1 \end{pmatrix}$$

Problem 1.6. Solve the following system of linear equations, using Jordan-Gaussian eliminations and Cramer's rule respectively.

$$x - 2y + z = 2$$

$$2x - 5y + 3z = 5$$

$$5x - 12y + 7z = 12$$

Are the column vectors linearly independent?

Problem 1.7. Let $A = \begin{pmatrix} 2 & -11 & -11 & -3 \\ -3 & 13 & 11 & 5 \\ 5 & -17 & -11 & -9 \\ -7 & 19 & 11 & 7 \end{pmatrix}$, $\mathbf{v}_1 = \begin{pmatrix} 6 \\ 7 \\ 8 \\ 9 \end{pmatrix}$, $\mathbf{v}_2 = \begin{pmatrix} 14 \\ 13 \\ 4 \\ 7 \end{pmatrix}$, and

$$\mathbf{b} = \begin{pmatrix} -180 \\ 206 \\ -258 \\ 242 \end{pmatrix}$$

It is known that $\mathbf{v}_1$ and $\mathbf{v}_2$ are two solutions to $A\mathbf{x} = \mathbf{b}$.

(1) Find another solution to $A\mathbf{x} = \mathbf{b}$.

(2) Given the reduced row echelon form R of A : $R = \begin{pmatrix} 1 & 0 & 0 & 4 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 0 \end{pmatrix}$, using this, or

otherwise, find the general solution to $A\mathbf{x} = \mathbf{b}$.

Problem 1.8. (Trace). For any $n \times n$ square matrix M , with the (i,j) -entry $m_{i,j}$, the trace of M is defined to be

$$\text{tr}(M) := \sum_{i=1}^n m_{ii}$$

Prove that:

(1) $\text{tr}(AB) = \text{tr}(BA)$ for $A, B \in \mathcal{M}_{n \times n}(\mathbb{R})$, and

(2) $\text{tr}(A^T B) = \text{tr}(AB^T)$ for $A, B \in \mathcal{M}_{m \times n}(\mathbb{R})$

(Note that in (2) we do not require them to be square matrices.)

Problem 1.9. (Optional)(Strictly diagonal dominant). Let A be a square matrix such that in every row, the diagonal element "dominates" the other entries. In other words,

$$|a_{i,i}| > \sum_{j \neq i} |a_{i,j}|, \forall i$$

Prove that A is invertible.

Solutions

Solution 1.1. 0.

Solution 1.2. 1. $\emptyset$.

2. ABC.

Solution 1.3. 1. False. To give a counter example, consider

$$A = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix}$$

2. False. $\begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix} \mathbf{x} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$ has unique solution.

3. True. $\det(A) \neq 0$ under both circumstances.

4. False. Consider $\mathbf{u} = 0$.

5. True. Replacing the last row with zeros, subtracting this from the original matrix. It is clear that these two are non-invertible.

6. False. To give a counter example, consider $\begin{pmatrix} 1 & 1 \\ 0 & 0 \\ 0 & 0 \end{pmatrix} \mathbf{x} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$

Solution 1.4. $(a, b, c) = (-1, 6, 5)$

Solution 1.5.

$$M^{-1} = \begin{pmatrix} \frac{1}{2} & -\frac{3}{2} & -\frac{1}{2} \\ \frac{3}{16} & -\frac{7}{16} & \frac{1}{16} \\ -\frac{1}{4} & \frac{5}{4} & \frac{1}{4} \end{pmatrix}$$

Remark. This actually comes from a midterm question several years ago, and in the original

question $\mathbf{M} = \begin{pmatrix} 3 & 4 & 5 & 6 \\ 1 & 0 & 2 & 2 \\ -2 & 4 & -1 & 0 \\ 1 & 2 & 1 & 1 \end{pmatrix}$, but I consider it quite time-consuming.

In case you would like to have a try, the answer is

$$\begin{pmatrix} \frac{1}{2} & -\frac{3}{2} & -\frac{1}{2} & 0 \\ -\frac{1}{8} & \frac{1}{8} & \frac{1}{8} & \frac{1}{2} \\ -\frac{3}{2} & \frac{5}{2} & 12 & 2 \\ \frac{5}{4} & -\frac{9}{4} & -\frac{1}{4} & -2 \end{pmatrix}$$

.

Solution 1.6. Gaussian eliminations:

The second equation becomes $y - z = -1$, and the third one gives you exactly the same. Hence we have $x = z$ and $y = z - 1$

Cramer's rule:

It simply doesn't work. The corresponding matrices are singular.

The column vectors (of the coefficient matrix) are not linearly independent. This can be easily seen doing to the rows are not, or, by observing that the second column is the inverse of the sum of the rest two columns.

Solution 1.7. (1) Another solution can be $\frac{(\mathbf{v}_1 + \mathbf{v}_2)}{2}$.

(2) As from the reduced row echelon form we can see that there is one free variable. Thus the solution set should be one-dimensional in $\mathbb{R}^4$. Since it is clear that any vector in the form of $a\mathbf{v}_1 + (1 - a)\mathbf{v}_2$ is a solution, and this set is one-dimensional in the space. The general solution is $\{a\mathbf{v}_1 + (1 - a)\mathbf{v}_2 | a \in \mathbb{R}\}$.

Solution 1.8. (1) $tr(AB) = \sum_{i=1}^n (AB)_{ii} = \sum_{i=1}^n \sum_{j=1}^n a_{ij}b_{ji}$

As each pair appears exactly once in the summation, and the pairing remains the same for BA , we may change the order of the summation.

That is, $tr(AB) = \sum_{i=1}^n \sum_{j=1}^m a_{ij}b_{ji} = \sum_{i=1}^n \sum_{j=1}^m b_{ij}a_{ji} = tr(BA)$

(2) As the (i,j) -entry of B^T is the (j,i) -entry of B , $tr(AB^T) = \sum_{i=1}^n \sum_{j=1}^n a_{ij}b_{ij}$

Similarly, we have $tr(A^T B) = \sum_{i=1}^m \sum_{j=1}^n a_{ji}b_{ji}$, which is just the same as $tr(AB^T)$

Solution 1.9. Here we use the fact that a square $n \times n$ matrix A is invertible when $Ax=0$ if and only if $x=0$. It is possible to derive this using the uniqueness of the solution to $Ax = 0$, but generally speaking, it's a property to be introduced later.

Proof by contradiction: First assume that for some $x \neq 0$ such that $Ax = 0$. Then $\sum_{j=1}^n a_{ij}x_j = 0, \forall i \in \{1, \dots, n\}$.

Take x_k to be the largest entry of x concerning absolute value, then $0 = \sum_{j=1}^n a_{kj}x_j$. By moving x_{kk} to the other side we obtain

$$a_{kk} = - \sum_{j \neq k} a_{kj} \frac{x_j}{x_k}$$

. By taking the absolute value and apply the triangle inequality, we get

$$\begin{aligned} |a_{kk}| &= \left| - \sum_{j \neq k} a_{kj} \frac{x_j}{x_k} \right| \\ &\leq \sum_{j \neq k} \left| a_{kj} \frac{x_j}{x_k} \right| \\ &= \sum_{j \neq k} |a_{kj}| \left| \frac{x_j}{x_k} \right| \\ &\leq \sum_{j \neq k} |a_{kj}| \end{aligned}$$

This contradicts the requirement of diagonal dominance. Thus A maps only zero element to zero, and hence it is invertible.

2 Vector Spaces

Problem 2.1. Let $v_1, v_2, \dots, v_n$ be a set of vectors in some vector space V , $w_k = \sum_{i=1}^k v_i$. Prove or disprove that

$$\text{span}(v_1, v_2, \dots, v_n) = \text{span}(w_1, w_2, \dots, w_n)$$

Problem 2.2. The solution set of the system of linear equations

$$x - 2y + z = 0$$

$$2x - 3y + z = 0$$

is a subspace of $\mathbb{R}^3$. (Verify it if the result is not obvious to you.) Find a basis for this subspace.

Problem 2.3. Given S and T are two subspaces of $\mathbb{R}^n$, show that $S \cup T$ is a subspace if and only if one of them is actually in the other.

Problem 2.4. (1) Prove that $\mathcal{M}_{n \times n}(\mathbb{R})$, the space of $n \times n$ square matrices with real entries, is a vector space.

(2) Prove that symmetric matrices ($Sym_n := \{A \in \mathcal{M}_{n \times n}(\mathbb{R}) | A = A^T\}$) and skew-symmetric matrices ($Skew_n := \{A \in \mathcal{M}_{n \times n}(\mathbb{R}) | A = -A^T\}$) are subspaces in $\mathcal{M}_{n \times n}(\mathbb{R})$.

(3) Find the dimension of the mentioned three spaces.

(4)(Optional) Prove that $\mathcal{M}_{n \times n}(\mathbb{R}) = Sym_n + Skew_n$

(The addition of spaces are given by $A + B = \{a + b | a \in A, b \in B\}$)

Problem 2.5. (Optional)(Dual Space) Verify that $\mathbb{R}^S := \{f | f \text{ is a function from } S \text{ to } \mathbb{R}\}$ is indeed a vector space with *sum* and *scalar product* defined as below:

$$(f + g)(x) = f(x) + g(x), \forall x \in S$$

$$\forall f, g \in \mathbb{R}^S (\lambda f)(x) = \lambda f(x), \forall \lambda \in \mathbb{R}, \forall f \in \mathbb{R}^S$$

If $S = \mathbb{R}^n$, then it is obvious that the space is infinite dimensional. However, if we consider only the linear functions, the space of linear functions is actually of dimension n . Can you give a basis for this space?

Hint: consider $f_i(e_j)$

Solutions

Solution 2.1. *Proof.* As $v_i = w_i - w_{i-1}$ for $i \geq 2$ and $v_1 = w_1$, for each $v \in \text{span}(v_1, v_2, \dots, v_n)$, $v = \sum_{i=1}^n c_i v_i$ for some $c_i \in \mathbb{R}$.

Then, $v = c_1 v_1 + \sum_{i=2}^n c_i (w_i - w_{i-1}) = \sum_{i=1}^n c'_i w_i$ for some $c'_i \in \mathbb{R}$. Hence $v \in \text{span}(w_1, w_2, \dots, w_n)$.

Then, $v = c_1 v_1 + \sum_{i=2}^n c_i (w_i - w_{i-1}) = \sum_{i=1}^n c'_i w_i$ for some $c'_i \in \mathbb{R}$. Hence $v \in \text{span}(w_1, w_2, \dots, w_n)$.

Then $\text{span}(v_1, v_2, \dots, v_n) \subset \text{span}(w_1, w_2, \dots, w_n)$.

The other direction is essentially the same.

Solution 2.2. Using any method, you will have $x = y = z$. The solution set is $\{x \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \mid x \in$

$\mathbb{R}\}$, or $\text{span} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ a basis can be $\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$

Solution 2.3. If: Obvious.

Only if: Assume the contrary, then $\exists s \in S, s \notin V$ and $\exists v \in V, v \notin S$.

As $S \cup T$ is a subspace and $v, s \in S \cup T$, so is $v + s$. If it's in S , then $v = (v + s) + (-s) \in S$, which is a contradiction. And similarly for V . Hence the assumption is false.

Solution 2.4. (1)&(2) The answer is obvious, but you are expected to verify each requirement respectively. Just do it entrywisely.

(3) As each entry is not "linearly dependent" of the others, we may guess that the space is of dimension n^2 . To verify it, we need to find a basis, which is rather simple. Take A_{ij} to be the matrix such that $a_{ij} = 1$ and all the other entries are zero. Then $\{A_{ij} \mid 1 \leq i, j \leq n\}$ forms a basis of $\mathcal{M}_{n \times n}(\mathbb{R})$. Note that in Sym_n , a matrix is uniquely determined by the upper half (or equivalently, the lower half) including the diagonal, and there is no further requirements regarding those entries, the space has dimension $\frac{n(n+1)}{2}$.

The case is essentially the same for Skew_n , except that the diagonal entries are required to be zero. Thus the dimension is $\frac{n(n-1)}{2}$.

(4) As you may observe, given any square matrix A , $\frac{A+A^T}{2} \in \text{Sym}_n$ and $\frac{A-A^T}{2} \in \text{Skew}_n$, which can be easily verified. This shows that $\mathcal{M}_{n \times n}(\mathbb{R}) = \text{Sym}_n + \text{Skew}_n$.

Further remark (More optional) In Linear Algebra II (2102), you will be introduced the notation of direct sum $\oplus$. The sum $A + B$ is direct if any element of $A \oplus B$ can be represented by some unique $a + b, a \in A \text{ and } b \in B$. You may interpret this by thinking that the two spaces have no overlap, but not in the intersection way.

In this example, we actually have $\mathcal{M}_{n \times n}(\mathbb{R}) = \text{Sym}_n \oplus \text{Skew}_n$. This can be also verified entrywisely, and may be a further exercise in case you are interested.

Solution 2.5. By verifying the conditions one by one, it should not be hard to see that the space is a vector space. This problem is designed to show that vectors can be more than a list of numbers. (In the previous question, matrices in this sense can be interpreted as a vector of $m \times n$ dimension.)

General function spaces can be studied with some norm or/and inner product. (A typical topic for Functional Analysis, Math 4404.) Linear functions, are especially interesting because they are related to linear transformation, or matrices, and also because they are related to derivatives.

Take $f_i(x)$ such that $f_i(e_j) = \delta_{ij}$. Here the Kronecker delta is defined to be $\delta_{ij} = 1, \text{ if } i = j$ and 0, otherwise. There are in total n such functions and the span of them is indeed the whole space of linear functions from $\mathbb{R}^n$ to $\mathbb{R}$.

3 Dimension and Basis

Problem 3.1. Which of the following subsets of $\mathbb{R}^3$ is indeed a subspace of it? Find a basis for those. (By default, $\mathbf{x} = (\mathbf{x}, \mathbf{y}, \mathbf{z})$)

- (a) $\{\mathbf{x} | \mathbf{x} = \mathbf{y} \text{ and } z = 0\}$.
- (b) $\{\mathbf{x} | \mathbf{x} = \mathbf{y} + \mathbf{1}\}$.
- (c) $\{\mathbf{x} | \mathbf{x}, \mathbf{y}, \mathbf{z} \text{ are positive}\}$.
- (d) $\{\mathbf{x} | \mathbf{x} - \mathbf{y} + \mathbf{z} = \mathbf{k}, \mathbf{k} \text{ is a constant}\}$

Problem 3.2. Find the dimension and a basis for the following subspace W of $\mathbb{R}^4$

$$W = \text{span}\{(1, -1, 0, 1)^T, (3, 0, 4, -2)^T, (0, 1, 2, -1)^T, (4, -2, 1, -1)^T\}$$

$$W = \{(x, y, z, w) \in \mathbb{R}^4 : 2x + 3y + 4z + 5w = 3x - y - 2z + w = 0\}$$

Problem 3.3. Let U and V be subspaces of some finite-dimensional vector space. Prove that

$$\dim(U + V) + \dim(U \cap V) = \dim U + \dim V$$

Hint: you may construct bases of the spaces mentioned.

Problem 3.4. All $n \times n$ matrices with trace 0 forms a subspace of $\mathcal{M}_{n \times n}(\mathbb{R})$. Find a basis, and hence find the dimension of it.

Problem 3.5. (Optional)(Quotient Space) Let Y be a subspace of a vector space X . Given any $x \in X$, define

$$x + Y := \{v | v = x + y \text{ for some } y \in Y\}$$

Also define the regular addition and scalar multiplication by

$$(a + Y) + (b + Y) = (a + b) + Y$$

$$a(x + Y) = ax + Y$$

These elements form a vector space X/Y (reads X quotient Y). (Check it.) If the X is finite-dimensional, show that

$$\dim(X/Y) = \dim X - \dim Y$$

Hint: Note that different x does not necessarily result in different $x + Y$. You may consider what kinds of x give the same set $x + Y$, then form the proof by extending a basis for Y to a basis for X .

Solutions

Solution 3.1. (a) Yes. A basis is $(1, 1, 0)^T$.

(b) No. As the zero vector is not in this set.

(c) No. Still, it does not contain zero.

(d) It depends. If $k \neq 0$, then it's not, as it does not contain zero. Otherwise it is, and a basis is $\{(1, 1, 0)^T, (0, 1, 1)^T\}$

Solution 3.2. (i) Since

$$5(0, 1, 2, -1)^T + (1, -1, 0, 1)^T - 3(3, 0, 4, -2)^T + 2(4, -2, 1, -1)^T = (0, 0, 0, 0)^T$$

they are linearly dependent. Considering only the first three vectors, it is not difficult to show that they are linearly independent. Then the dimension is 3 and a basis can be $\{(0, 1, 2, -1)^T, (1, -1, 0, 1)^T, (3, 0, 4, -2)^T\}$.

(ii) Clearly W has dimension 2 and a basis can be $(2, -16, 11, 0)^T, (0, 7, -4, -1)^T$

Solution 3.3. First, as $U \cap V$ is a subspace, we may consider a basis $\{e_1, \dots, e_k\}$ for it. Then, it can be extended to a basis of U by adding some $\{u_1, \dots, u_p\}$, and similarly for V with $\{v_1, \dots, v_q\}$. Then, the union of the three sets is a basis for $U + V$.

The proof is simple. First, $\forall x \in U + V$, $x = u + v$, for some $u \in U, v \in V$. We have also

$$\begin{aligned} u &= \sum_{i=1}^k c_i e_i + \sum_{i=1}^p p_i u_i \\ v &= \sum_{i=1}^k c'_i e_i + \sum_{i=1}^q q_i v_i \\ x &= \sum_{i=1}^k (c_i + c'_i) e_i + \sum_{i=1}^p p_i u_i + \sum_{i=1}^q q_i v_i \end{aligned}$$

This suggests that it spans $U + V$

It remains to show that $\{u_1, \dots, u_p\}$ and $\{v_1, \dots, v_q\}$ are linearly independent. This is true, as if v_i can be represented by some linear combination of u_i s, it will be in U , but we have assumed in the very beginning that it's not.

Eventually, we may find the dimensions of the mentioned spaces, which are $(k + p + q)$, k , $k + p$, $k + q$, respectively.

$(k + p + q) + k = (k + p) + (k + q)$ is clearly true.

Solution 3.4. In **Problem 2.4** we have shown that

$$\{E_{i,j} | a_{i,j} = 1 \text{ and all other entries are } 0\}$$

form a basis for $\mathcal{M}_{n \times n}(\mathbb{R})$. By removing $\{E_{n,n}\}$ we have a basis for the required subspace. The dimension is naturally $n^2 - 1$.

Solution 3.5. (This problem is actually simpler than **Problem 3.3**, but the concept is a little bit abstract.)

First, note that $x_1 + Y = x_2 + Y$ if $(x_1 - x_2) \in Y$.

For example, if we take $\mathbb{R}^3/\mathbb{R}^2$ (or more formally, the plane $z = 0$), then it's like cutting the 3-dimensional spaces into slices. Clearly we have $(0, 0, 1)^T + \mathbb{R}^2 = (1, 1, 1)^T + \mathbb{R}^2$, as they give the same plane $z = 1$.

Returning to our problem, we first find a basis for Y , $\{e_1, \dots, e_k\}$, and by adding $\{e_{k+1}, \dots, e_n\}$ it becomes a basis for X . We claim that $\{e_{k+1}, \dots, e_n\}$ is a basis for X/Y . The linear independence is clear. Now we show that it spans X/Y .

Given any $x \in X$, $x = \sum_{i=1}^n c_i e_i$. Consider $x' = \sum_{i=k+1}^n c_i e_i$. We have $x + Y = x' + Y$, as $x - x' = \sum_{i=1}^k c_i e_i \in Y$. This suggests that any element in X/Y can be represented by some linear combination of $\{e_{k+1}, \dots, e_n\}$.

Then, it's clear that $\dim(X/Y) = \dim X - \dim Y$.

4 Linear Transformation

Problem 4.1. Given $T \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x+y \\ 1 \end{pmatrix}$, determine if T is a linear transformation.

Problem 4.2. Define $S : \mathbb{R}^3 \rightarrow \mathbb{R}^4$ and $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ respectively by

$$T(x_1, x_2, x_3) = (x_1 + 3x_2 - x_3, x_1 + 2x_3, x_2 + 4x_3)$$

$$S(x_1, x_2, x_3) = (x_2 + x_3, x_3 + x_1, x_1 + x_2, x_1 + x_2 + x_3)$$

Find $(S \circ T)(x_1, x_2, x_3)$, and hence give the matrix representation with respect to the standard basis.

Problem 4.3. (a) Let $\mathcal{P}$ represent any parallelogram (including interior points) in $\mathbb{R}^2$ that has the origin as a vertex, and that $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be an invertible linear transformation. Show that the image set $T(\mathcal{P})$ is also a parallelogram with one vertex at the origin.

(b) Also, let $\det(T)$ be the determinant of the representing matrix of T and $S(\cdot)$ be the default area of the input graph in the plane $\mathbb{R}^2$, show that

$$\det(T) = \frac{S(T(\mathcal{P}))}{S(\mathcal{P})}$$

(C) State without proof a three-dimensional result for (a) and (b).

Problem 4.4. Show that all linear transformations from one finite-dimensional vector space to another form a vector space. Without loss of generality, let the two spaces be $\mathbb{R}^m$ and $\mathbb{R}^n$. Find a basis and hence the dimension of this space.

Problem 4.5. (Optional)(Bilinear form)

(a) A bilinear form is in the form of $T : (x, y) \rightarrow \mathbb{R}$ for some vectors $x \in X$ and $y \in Y$, such that T is linear in each slot. That is, for any $a, b \in \mathbb{R}$ and $x, x_1, x_2 \in X, y, y_1, y_2 \in Y$, there is

$$T(x, ay_1 + by_2) = aT(x, y_1) + bT(x, y_2)$$

$$T(ax_1 + bx_2, y) = aT(x_1, y) + bT(x_2, y)$$

Show that in general, a bilinear form is not linear.

(b) Let X and Y be $\mathbb{R}^m$ and $\mathbb{R}^n$ respectively, define $T_x : Y \rightarrow \mathbb{R}$, $T_x(y) = T(x, y)$ for $x \in X$. Show that T_x forms a vector space.

(c) Now, further define S such that $S(x) = T_x$. Show that S is a linear transformation.

Problem 4.6. Show that a linear transformation T is injective if and only if $\text{Null}(T) = \{0\}$ (I know it's from the lecture notes.)

Problem 4.7. Show that two finite dimensional vector spaces are isomorphic if and only if they have the same dimension.

Remark. Two spaces are isomorphic if there exists an isomorphism from one to the other.

Problem 4.8. (Graph) Given a linear transformation T from X to Y , define the graph of T to be:

$$\Gamma(T) := \left\{ \begin{pmatrix} x \\ y \end{pmatrix} \mid x \in X, y = T(x) \right\}$$

Show that $\Gamma(T)$ is a bijective linear transformation.

Problem 4.9. (Isometry) A linear transformation from a normed vector space to another is said to be isometric if:

$$\|Tv\| = \|v\|, \forall v \in D(T)$$

In case you are not familiar with the term norm, it can be viewed as some distance from a point to the origin. The default norm on $\mathbb{R}^n$ is $\|x\| = \sqrt{\sum_{i=1}^n x_i^2}$, for $x = (x_1, x_2, \dots, x_n)^T$.

(a) Show that an isometry (or equivalently, an isometric linear transformation) must be injective.

(b) Show that if T is an isometry from a subspace W of $\mathbb{R}^n$ to W , it can be extended to some isometry defined on the whole space $\mathbb{R}^n$

Problem 4.10. (Optional)(Canonical mapping) Recall that in **Problem 2.5** we introduced the space of linear functions from $\mathbb{R}^n$ to $\mathbb{R}$. It is actually a vector space and has dimension n . We generally note the space of linear functions from X to $\mathbb{R}$ by X' . Then, it will be interesting to consider what is $(\mathbb{R}^n)''$.

Define the Canonical mapping to be:

$$\mathbf{C}(x) = g_x, g_x \in (\mathbb{R}^n)''$$

Here $g_x(f) = f(x), \forall f \in (\mathbb{R}^n)'$, or equivalently, f is linear and $f(x) \in \mathbb{R}$.

Show that $\mathbf{C}$ is linear and bijective, and hence $(\mathbb{R}^n)'' \cong \mathbb{R}^n$ ($\cong$ means *is isomorphic to*).

Hint: Here $\mathbf{C}(x)$ is changing a point into some kind of action, which inserts this point back into all linear functions and gets a real number. Don't be scared by g_x , just follow the definition of linearity and bijectivity. (Note that two functions are equal if and only if they have the same value for all inputs.)

Solutions

Solution 4.1. No. A necessary condition is that T maps 0 to 0. Here this is not the case.

Solution 4.2. (I made a mistake in the first line, clearly $R(S) = \mathbb{R}^4$ and $R(T) = \mathbb{R}^3$)
 $(S \circ T)(x_1, x_2, x_3) = (x_1 + x_2 + 6x_3, x_1 + 4x_2 + 3x_3, 2x_1 + 3x_2 + x_3, 2x_1 + 4x_2 + 5x_3).$

$$S \circ T = \begin{pmatrix} 1 & 1 & 6 \\ 1 & 4 & 3 \\ 2 & 3 & 1 \\ 2 & 4 & 5 \end{pmatrix}$$

Solution 4.3. (a) A parallelogram with a vertex the origin is uniquely determined by two vertexes such that the third vertex is the vector sum of the two. For example, in OABC we have $B = A + C$.

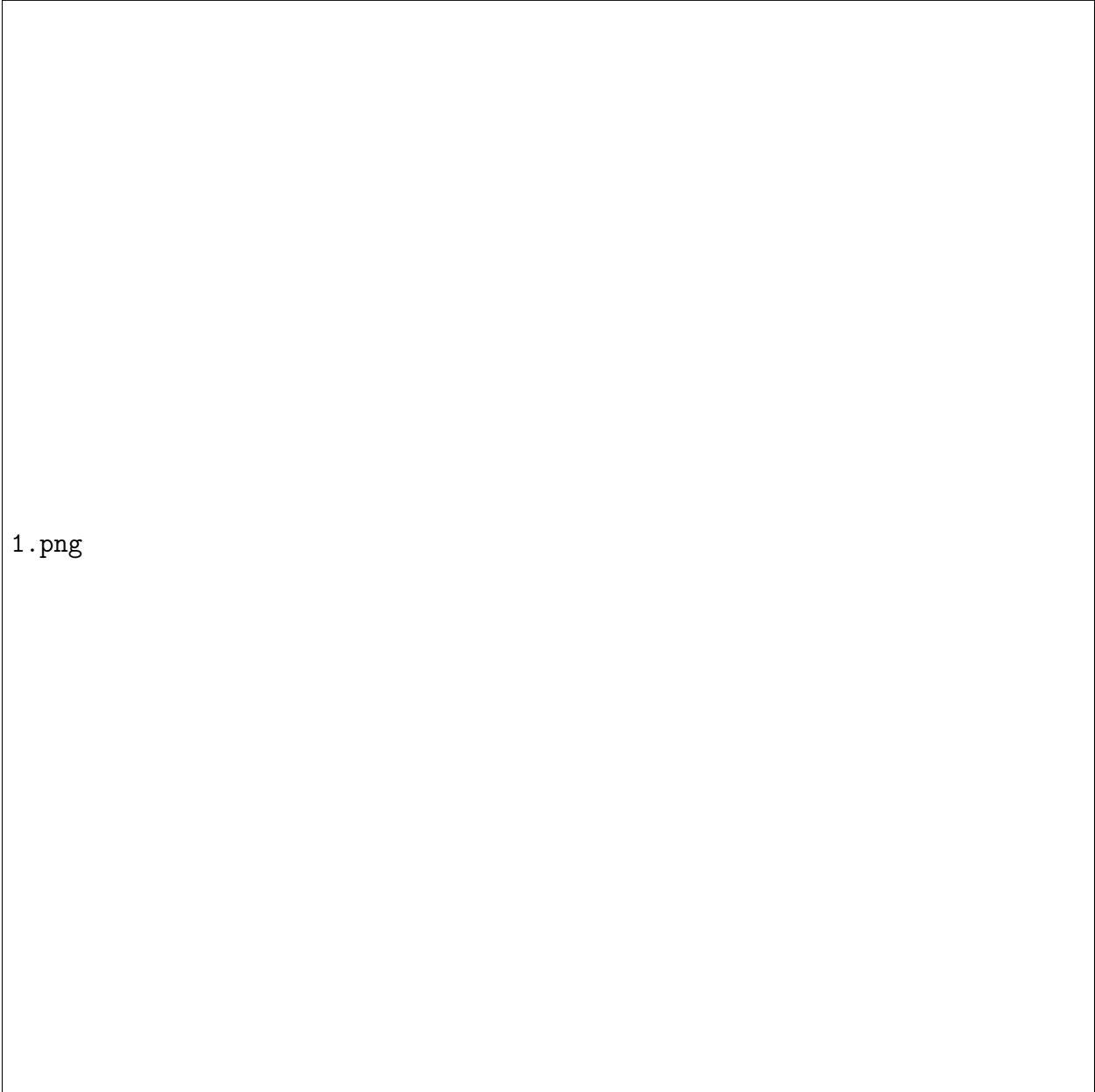
We first consider only the vertexes. As T is linear, $T(B) = T(A + C) = T(A) + T(C)$. And since T is invertible, it cannot map two non-parallel vectors to parallel ones. Hence, $OT(A)T(B)T(C)$ is indeed the vertexes of a parallelogram.

For any point x between AB, $x = A + \lambda(B - A)$ for some $\lambda \in (0, 1)$. Then $T(x) = T(A) + T(\lambda(B - A)) = T(A) + \lambda(T(B) - T(A))$ lies at the same position between $T(A)T(B)$. As all points on the sides lie between two vertexes, and all points inside the parallelogram lie between some two points on the sides, the image set is also a parallelogram.

(b) We consider first the unit square $[0, 1] \times [0, 1]$ in $\mathbb{R}^2$. Given T has the matrix representation $\begin{pmatrix} a & c \\ b & d \end{pmatrix}$ with $x_1y_2 - y_1x_2 \neq 0$ as T is invertible, the image of the unit square is given by $\begin{pmatrix} a \\ b \end{pmatrix}$ and $\begin{pmatrix} c \\ d \end{pmatrix}$ being $OT(A)$ and $OT(C)$ respectively.

Consider the following picture that gives the proof, though more vigorously, we have to take those circumstances where some or all of a, b, c, d is/are negative into account, the actual result remains unchanged.

We may start from an arbitrary parallelogram and use a same method to get the same result, but the computation is really complicated. This question should be much easier if you have learned external product and can calculate the area of a parallelogram using it. (May I assume that you have all learned it in high school?)



1.png

Solution 4.4. This is an easy question. Any such linear transformation has a matrix presentation. Then, it's just the space $\mathcal{M}_{n \times m}(\mathbb{R})$. Clearly, as we've seen before, the dimension is mn .

Solution 4.5. (a) $T(ax, ay) = aT(x, ay) = a^2T(x, y) \neq aT(x, y)$ in general. Then T is not linear.

(b) From the previous question, since T_x is linear of y , it's a subset of $\mathbf{B}(Y, \mathbb{R})$, linear transformations from Y to $\mathbb{R}$. It remains to show that it's a subspace.

First, we show T_0 is 0. This is clear as $T_0(y) = T(0, y) = T(0 + 0, y) = 2T(0, y)$.

Then, $T_{x_1}(y) + T_{x_2}(y) = T(x_1, y) + T(x_2, y) = T(x_1 + x_2, y) = T_{x_1+x_2}(y)$ shows that it's closed under addition.

$T_{ax}(y) = T(ax, y) = aT(x, y) = aT_x(y)$ shows that it's closed under scalar multiplication.

Hence, $\{T_x | x \in X\}$ forms a vector space.

(c) This is just the same as in (b).

Solution 4.6. If: Given $\text{Null}(T) = \{0\}$, then $T(x) = T(y) \iff T(x - y) = 0$ gives $x - y = 0$. Hence $x = y$, and this suggests T is injective.

Only if: Since T is injective, $T(x) = 0$ implies $x = 0$, then the result is clear.

Solution 4.7. Since they are of the same dimension, let $\{e_1, \dots, e_n\}$ and $\{e'_1, \dots, e'_n\}$ be two bases for the spaces. Then define T such that $T(e_i) = e'_i$. T is clearly linear, bijective and thus isomorphic. If we consider orthonormal bases, which is to be introduced in the next chapter, we will see that it's still an isomorphism for normed and even inner-product spaces.

Solution 4.8. First we show linearity by verifying that:

$\Gamma(T)(0) = 0$, as $T(0) = 0$

$$\Gamma(T)(x_1 + x_2) = \begin{pmatrix} x_1 + x_2 \\ T(x_1) + T(x_2) \end{pmatrix} = \begin{pmatrix} x_1 \\ T(x_1) \end{pmatrix} + \begin{pmatrix} x_2 \\ T(x_2) \end{pmatrix} = \Gamma(T)(x_1) + \Gamma(T)(x_2)$$

$$\Gamma(T)(ax) = \begin{pmatrix} ax \\ T(ax) \end{pmatrix} = a \begin{pmatrix} x \\ T(x) \end{pmatrix} = a\Gamma(T)(x)$$

Observe that the first slot of $\Gamma(T)$ is x , which gives injectivity, while the surjectivity is clear because the graph is defined to be the whole range.

Solution 4.9. (a) Let $T(x) = T(y)$ for some x, y and isometric T .

Then $\|x - y\| = \|T(x - y)\| = \|T(x) - T(y)\| = 0$. This gives $x = y$. Hence T is injective.

(b) Consider a basis for W , $\{e_1, \dots, e_m\}$. We may extend it to the whole space $\mathbb{R}^n$ into $\{e_1, \dots, e_n\}$.

As this is indeed a basis, any $x \in \mathbb{R}^n$ can be represented uniquely by some linear combination of the vectors, i.e. $x = \sum_{i=1}^n c_i e_i$.

Then, x can be uniquely represented by $w + v$, where $w \in W$ and $v \in \text{span}\{e_{m+1}, \dots, e_n\}$. (It is actually the $\oplus$ we did not go further to introduce in **Problem 2.4**.)

Define $\tilde{T}(v+w) := T(v) + w$, where v and w are defined as previously. Then $\tilde{T}$ is an isometry.

We verify this in the following procedure. First, we have that $T(\sum_{i=1}^m c_i e_i) = \sum_{i=1}^m k_i e_i$ such that $\sum_{i=1}^m c_i^2 = \sum_{i=1}^m k_i^2$. This is a result from that T is an isometry in W . Given any $x \in \mathbb{R}^n$, $x = \sum_{i=1}^n c_i e_i$. Then $\|x\|^2 = \sum_{i=1}^n c_i^2$.

Now we calculated the norm of $\tilde{T}(x)$:

$$\begin{aligned} \|\tilde{T}(x)\|^2 &= \|\tilde{T}(\sum_{i=1}^n c_i e_i)\|^2 = \|T(\sum_{i=1}^m c_i e_i) + (\sum_{i=m+1}^n c_i e_i)\|^2 = \|\sum_{i=1}^m k_i e_i + \sum_{i=m+1}^n c_i e_i\|^2 \\ &= \sum_{i=1}^m k_i^2 + \sum_{i=m+1}^n c_i^2 = \sum_{i=1}^m c_i^2 + \sum_{i=m+1}^n c_i^2 = \sum_{i=1}^n c_i^2 \end{aligned}$$

This gives that $\tilde{T}$ is isometric.

Solution 4.10. Firstly, we show the linearity of $\mathbf{C}$.

$\mathbf{C}(0) = 0$ as $g_0(f) = f(0) = 0, \forall f \in (\mathbb{R}^n)'$.

$\mathbf{C}(ax)(f) = g_{ax}(f) = f(ax) = af(x) = ag_x(f) = a\mathbf{C}(x)(f)$.

$\mathbf{C}(x+y)(f) = g_{x+y}(f) = f(x+y) = f(x) + f(y) = (g_x + g_y)(f) = (\mathbf{C}(x) + \mathbf{C}(y))(f)$.

Secondly, we show that $\mathbf{C}$ is injective. Here we let $x = (x_1, \dots, x_n) \in \mathbb{R}^n$.

$\forall x \neq 0, \exists x_i \neq 0$. Define $\pi_i(x) = x_i$ to be the projection on the i -th dimension, then it's clearly linear. Then $g_x(\pi_i) = \pi_i(x) \neq 0$ shows that $\mathbf{C}(x) \neq 0, \forall x \neq 0$. This gives injectivity.

Thirdly, we show that $\mathbf{C}$ is surjective. Here surjectivity suggests that $\mathbf{C}$ gives all the possible ways to getting f back to some c in $\mathbb{R}$

Given any linear functional f . If $f = 0$, then it's already shown that $\mathbf{C}(0) = 0$. Otherwise $\exists x \in \mathbb{R}^n$ such that $f(x) = c' \neq 0$. (As f is non-zero)

Then, by letting $x' = \frac{c}{c'}x$, $\mathbf{C}(x')(f) = g_{x'}(f) = f(x') = \frac{c}{c'}f(x) = c$.

(The previous line has been changed somehow. Please refer to this as the final version instead of the recording. Thanks.)

Hence, any functional from $f \in (\mathbb{R}^n)'$ to c' can be obtained by $\mathbf{C}(x')$. Hence $\mathbf{C}$ is surjective.

As $\mathbf{C}$ is bijective, it's an isomorphism. Then $(\mathbb{R}^n)'' \cong \mathbb{R}^n$

A tricky way of doing so is that: As the space all have dimension n , which is shown in **Problem 4.4**. $\dim((\mathbb{R}^n)') = \dim(\mathbb{R}^n) \times \dim(\mathbb{R}) = n$, and so for $(\mathbb{R}^n)''$.

Thus if $\mathbf{C}$ is injective, $R(\mathbf{C}) \in (\mathbb{R}^n)''$, and since they are of the same dimension, they must be equal. Hence $\mathbf{C}$ is surjective. Similarly, If it's surjective, by $D(\mathbf{C}) \in \mathbb{R}^n$ we get the surjectivity. You may choose whichever direction you think is easier.

Also, the isomorphism is a direct result that two vector spaces of the same dimensions are always isomorphic.

5 Change of Basis and Similarity

Problem 5.1. Compute the matrix representation $[Id]_{\beta_1}^{\beta_2}$ of the change from $\beta_1 = \{(1, 1, 1)^T, (1, 2, 5)^T, (0, -1, 4)^T\}$ back to the standard basis $\beta_2 = \{e_1, e_2, e_3\}$ of $\mathbb{R}^3$.

Problem 5.2. Let $\beta = \{(2, 0, 1, 1)^T, (2, 1, 3, 1)^T, (1, 1, 3, 1)^T, (1, 1, 1, 0)^T\}$ be an ordered basis of $\mathbb{R}^4$, and T be the linear map such that $T(a, b, c, d)^T = (a - b, b - c, c - d, d - a)^T$. Find the matrix representation $[T]_{\beta}^{\beta}$.

Problem 5.3. Show that matrix similarity is an equivalence relation. (In case you have not taken Math2012, a relation $\sim$ is an equivalence relation if: (i) $a \sim a$ for any a ; (ii) $a \sim b$ gives $b \sim a$; and (iii) $a \sim b$ and $b \sim c$ gives $a \sim c$.)

Problem 5.4. Similar matrices have quite a lot of common features. Verify that they have the same (i) Trace; (ii) Determinant; and (iii) Rank.

Problem 5.5. Define T to be the conjugation action by A on $\mathcal{M}_{2 \times 2}(\mathbb{R})$:

$$M \mapsto A^{-1}MA$$

Now, given $A = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$, find $[T]_{\beta}$ given the standard basis β .

$$\beta = \left\{ \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \right\}$$

Solutions

Solution 5.1. $\begin{pmatrix} 1 & 1 & 0 \\ 1 & 2 & -1 \\ 1 & 5 & 4 \end{pmatrix}$

Solution 5.2. (The answer has been corrected. Apologies for the previous mistake.)

$$[T]_{\beta}^{\beta} = \begin{pmatrix} -3 & -6 & -4 & -3 \\ 9 & 15 & 10 & 6 \\ -7 & -10 & -6 & -4 \\ -3 & -7 & -6 & -2 \end{pmatrix}$$

This can be obtained by calculating $[Id]_{\alpha}^{\beta}[T]_{\alpha}^{\alpha}[Id]_{\beta}^{\alpha}$ (α stands for the standard basis of $\mathbb{R}^4$), or calculating $T\beta_i$ and rewrite with respect to β .

Solution 5.3. We use $\sim$ to represent ‘similar to’ in this and the following question.

Clearly $M \sim M$, as $M = I^{-1}MI$;

If $A \sim B$, $A = P^{-1}BP$. Then $B = PP^{-1}BPP^{-1} = PAP^{-1} = (P^{-1})^{-1}AP^{-1}$, which gives $B \sim A$;

If $A \sim B$, $B \sim C$, then $A = P^{-1}BP = P^{-1}Q^{-1}CQP = (QP)^{-1}C(QP)$, which gives $A \sim C$.

Solution 5.4. Let $A \sim B$.

(i) In **Problem 1.8** we have shown that for any square matrices A, B , $tr(AB) = tr(BA)$. Then $tr(A) = tr(P^{-1}BP) = tr(P^{-1}PB) = tr(IB) = tr(B)$.

(ii) We have $det(AB) = det(BA)$, thus the same proof holds.

(iii) We show that for any $A : \mathbb{R}^n \rightarrow \mathbb{R}^n$ and any invertible M (such that the following things are well-defined), $rank(MA) = rank(A) = rank(AM)$.

Since M is invertible, it's surjective. Thus if $\{Av_1, \dots, Av_m\}$ forms a basis for $R(A)$, there exists $v'_1, \dots, v'_m$ such that $Mv'_i = Av_i$. And hence $dim(R(MA)) \geq dim(R(A))$; On the other hand, as $R(M) \subset \mathbb{R}^n$, $R(MA) \subset R(A)$. This gives $rank(MA) = rank(A)$.

Similarly, $dim(R(AM)) \leq dim(R(A))$. If $\{Av_1, \dots, Av_m\}$ forms a basis for $R(A)$, $\{AMv_1, \dots, AMv_m\}$ are also linearly independent, as M maps linearly independent sets to linearly independent sets. Then $dim(R(AM)) \geq dim(R(A))$. That is, $rank(AM) = rank(A)$

Solution 5.5. $[T]_{\beta} = \begin{pmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 \\ 1 & 1 & -1 & -1 \\ 1 & 0 & -1 & 0 \end{pmatrix}$

(Just insert the basis into it.)

6 Diagonalization

Problem 6.1. Find all eigenvalues of:

(a) $\begin{pmatrix} 2 & 3 \\ 1 & 6 \end{pmatrix}$

(b) $\begin{pmatrix} 0 & 5 & 1 \\ 1 & 4 & 7 \\ 0 & 0 & 0 \end{pmatrix}$

(c) $\begin{pmatrix} 0 & 0 & 1 & 1 \\ -2 & 2 & 1 & 1 \\ 7 & -1 & 1 & -2 \\ -13 & 1 & 4 & 7 \end{pmatrix}$

Problem 6.2. Provide an example of a linear transformation T defined on some finite-dimensional vector space such that T^2 is diagonalizable but T itself is not. (Here T^n represents applying T for n times.)

Remark. If T is invertible and the space is complex, then T^k is diagonalizable $\iff T$ is. We shall not go further to discuss that.

Problem 6.3. At the end of the lecture note 5, we've defined e^A . Now, given diagonalizable A , show that

$$\det(e^A) = e^{(\text{tr}(A))}$$

Problem 6.4. The *Fibonacci sequence* is defined to as:

$$F_0 = 0, F_1 = 1 \text{ and } F_n = F_{n-1} + F_{n-2} \text{ for } n \geq 2$$

Define $T \in \mathcal{L}(\mathbb{R}^2)$ (the linear transformations from $\mathbb{R}^2$ to $\mathbb{R}^2$, we normally call such T an operator) by $T(x, y) = (y, x + y)$.

- (a) Show that $T^n(0, 1) = (F_n, F_{n+1})$.
- (b) Find the eigenvalues of T .
- (c) Find a basis of $\mathbb{R}^2$ consisting of eigenvectors of T .
- (d) Use (c) to compute $T^n(0, 1)$, and conclude that

$$F_n = \frac{1}{\sqrt{5}} \left[\left(\frac{1 + \sqrt{5}}{2} \right)^n - \left(\frac{1 - \sqrt{5}}{2} \right)^n \right]$$

Problem 6.5. (*Optional*)(Polynomial of operator)

In the previous questions we have shown what T^n is. Given a monic polynomial $p(x) = x^n + a_{n-1}x^{n-1} + \dots + a_1x + a_0$, for simplicity we assume all coefficients and all roots of it to be real. It is natural to define $p(T)$, which is still a linear operator such that

$$p(T)(x) = T^n(x) + a_{n-1}T^{n-1}x + \dots + a_1T(x) + a_0x$$

- (a) Show that there exist a unique polynomial p (depending on T) of the lowest degree such that $p(T) = 0$. We call it the minimal polynomial of T .
- (b) Show that the roots of p are eigenvalues of T .

Solutions

Solution 6.1. (a) $4 + \sqrt{7}$ and $4 - \sqrt{7}$;
 (b) 5, -1, and 0;
 (c) 2 and 3.

Solution 6.2. A most simple example is $T = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$. It's not diagonalizable but $T^2 = O$ is, although trivially.

Solution 6.3. Since A is diagonalizable, we write $A = P^{-1}DP$ for some diagonal matrix D . Then clearly $A^n = P^{-1}D^nP$.

$$(e^A) = \sum_{n=1}^{\infty} \frac{1}{n!} P^{-1} D^n P = P^{-1} \left(\sum_{n=1}^{\infty} \frac{1}{n!} D^n \right) P = P^{-1} e^D P$$

Since D is diagonal, let $\{d_i\}$ be the elements on the diagonal of D , $e^D = \text{diag}(e^{d_1}, \dots, e^{d_n})$. As A is similar to D , they share the same trace (as we've show in the previsou section). Then, $\det(e^A) = \det(P^{-1})\det(e^D)\det(P) = \prod_{i=1}^n e^{d_i} = e^{\text{tr}(D)} = e^{\text{tr}(A)}$

Solution 6.4. (a) We use induction on n. Clearly $T^0(0, 1) = (0, 1) = (F_0, F_1)$. Given $T^k(0, 1) = (F_k, F_{k+1})$, $T^{k+1}(0, 1) = T(F_k, F_{k+1}) = (F_{k+1}, F_k + F_{k+1}) = (F_{k+1}, F_{k+2})$. This gives the result.

(b) Either find that $T = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$, or solving directly $Tv = \lambda v$ will gives that the eigenvalues are $\lambda_1 = \frac{1+\sqrt{5}}{2}$ and $\lambda_2 = \frac{1-\sqrt{5}}{2}$.

(c) Using the result of (b), the eigenvectors are $v_1 = (\frac{-1+\sqrt{5}}{2}, 1)$ and $v_2 = (\frac{-1-\sqrt{5}}{2}, 1)$. As they are linearly independent, they form a basis of $\mathbb{R}^2$.

(d) First we find that $(0, 1) = \frac{5+\sqrt{5}}{10}v_1 + \frac{5-\sqrt{5}}{10}v_2$.

Then, $T^n(0, 1) = T^n(\frac{5+\sqrt{5}}{10}v_1 + \frac{5-\sqrt{5}}{10}v_2) = \frac{5+\sqrt{5}}{10}\lambda_1^n v_1 + \frac{5-\sqrt{5}}{10}\lambda_2^n v_2$.

Focus only on y (since the y-coordinates of the eigenvectors are simpler),

$$F_{n+1} = \frac{5+\sqrt{5}}{10} \left(\frac{1+\sqrt{5}}{2} \right)^n + \frac{5-\sqrt{5}}{10} \left(\frac{1-\sqrt{5}}{2} \right)^n = \frac{1}{\sqrt{5}} \left(\frac{1+\sqrt{5}}{2} \right)^{n+1} - \frac{1}{\sqrt{5}} \left(\frac{1-\sqrt{5}}{2} \right)^{n+1}.$$

Which is, $F_n = \frac{1}{\sqrt{5}} \left[\left(\frac{1+\sqrt{5}}{2} \right)^n - \left(\frac{1-\sqrt{5}}{2} \right)^n \right]$

Solution 6.5. (a) Note that the linear combination of linear transformations is still a linear transformation. Then a polynomial of a linear transformation is still one.

As the multiplication of polynomials is commutative (i.e. $pq = qp$), the multiplication of polynomials of linear transformations is also commutative. The proof is not difficult and really intuitive. We omit it here.

We use induction on the dimension of the space V . If $\dim V$ is 1, then $Tx = cx$ for some constant c , then $p(x) = x - c$ will be the minimal polynomial of T , as $p(T)(v) = (T - c)v = Tv - cv = 0$ for any v .

Now assume that the statement is true for all spaces with a lower dimension than V . $\forall v \in V$, we may find a list of vectors: $\{v, Tv, \dots, T^{\dim V} v\}$, which contains exactly $(\dim V + 1)$ vectors, and hence must be linearly dependent. Assume that the list first becomes linearly dependent at some $T^m v$, with $1 \leq m \leq \dim V$.

Then, $c_0 v + c_1 Tv + \dots + c_{m-1} T^{m-1} v + T^m v = 0$ for some constants. We define $q(x) = c_0 + c_1 x + \dots + c_{m-1} x^{m-1} + x^m$. For any $k \geq 0$, $q(T)(T^k v) = T^k(q(T)v) = T^k(0) = 0$, by commutativity. As we have $\{v, Tv, \dots, T^{m-1} v\}$ are in $\text{null}(q(T))$ and are linearly independent, the null space of $q(T)$ has dimension larger or equal to m .

Now, as the dimension of $\text{null}(q(T))$ and $\text{range}(q(T))$ adds to $\dim V$, $\text{range}(q(T)) \leq \dim V - m$, thus by induction assumption, there exist some polynomial $r(x)$ such that $r(T)$ is the minimal polynomial for $\text{range}(q(T))$. Then, if we multiply them together, we have $(qr)(x)$ is the minimal polynomial for T . As for any v , $q(T)(v)$ lies in $\text{range}(q(T))$ and $r(T)$ sends all vectors in $\text{range}(q(T))$ to 0.

For uniqueness, if p and p' are both minimal polynomials, $(p - p')(T)(v) = 0$, and $p - p'$ has a lower degree than p and p' , which contradict that they are minimal polynomials.

Examples

(i) If T is the identity, then $p(x) = x - 1$ is the minimal polynomial for T , as $Tv - v = 0$ for all v .

(ii) If T is the rotation in $\mathbb{R}^2$ (i.e. $T = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$), then $p(x) = x^2 + 1$ is the minimal polynomial for T , as $T^2 v - v = 0$ for all v .

(iii) If T is the projection from $\mathbb{R}^3 \rightarrow \mathbb{R}^2$ (i.e. $T = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$), then $p(x) = x^2 - x$ is the

minimal polynomial for T , as $T^2 v - Tv = 0$ for all v .

(b) Since we assume p has all real roots, we may have $p(x) = (x - \lambda)q(x)$. Then $0 = p(T) = (T - \lambda I)q(T)$.

Because p is the minimal polynomial but not q , then there exists some v such that $q(T)(v)$ is non-zero but $p(T)(v)$ is. Then $(T - \lambda I)(q(T)v) = 0$ gives that λ is an eigenvalue of T . (Note that $q(T)(v)$ is a vector in V .)

On the other hand, if λ is a eigenvalue of T , $T(v) = \lambda v$, then $p(T)(v) = p(\lambda)(v) = 0$. Note that $p(\lambda)(v)$ is actually a scalar product instead of a linear transformation. This gives $p(\lambda) = 0$, then λ is a root of p .

7 Orthogonality

Problem 7.1. Check if the following is an orthogonal set:

$$\left\{ \begin{pmatrix} 1 \\ -4 \\ 3 \end{pmatrix}, \begin{pmatrix} 5 \\ 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 3 \\ -4 \\ -7 \end{pmatrix} \right\}$$

Problem 7.2. In previous sections, we've defined isometry as: a linear transformation from a vector space to itself such that the norm is preserved.

Show that an isometry also preserves orthogonality. That is, if $\langle x, y \rangle = 0$, so is $\langle T(x), T(y) \rangle$.

Problem 7.3. (a) Find an orthonormal basis of $\text{null} \begin{pmatrix} -1 & 2 & 3 & 1 \\ 2 & -4 & 5 & -1 \end{pmatrix}$.

(b) Calculate the projection of $\begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}$ onto this null space.

Problem 7.4. (*Optional*)

In previous sections, we've verified that polynomials defined on $[0, 1]$ with real coefficients form a vector space. Then, $\{1, x, x^2, x^3\}$ spans a subspace $W = \{ax^3 + bx^2 + cx + d | a, b, c, d \in \mathbb{R}\}$. In this space, an inner product can be defined by:

$$\langle x, y \rangle = \int_0^1 x(t)y(t)dt$$

It is known that it satisfies the requirements for an inner product (which is beyond this course). Use the Gram-Schmidt process to find an orthonormal basis of W .

Solutions

Solution 7.1. The set is NOT orthogonal, as $\left\langle \begin{pmatrix} 1 \\ -4 \\ 3 \end{pmatrix}, \begin{pmatrix} 3 \\ -4 \\ -7 \end{pmatrix} \right\rangle = 3 + 16 - 21 = -2 \neq 0$

Solution 7.2. Since T is isometric, $\|T(x - y)\| = \|x - y\|$. That is, $\langle T(x - y), T(x - y) \rangle = \langle x - y, x - y \rangle$.

Using the linearity of T and the properties of inner product, it is equivalent to:

$$\langle T(x), T(x) \rangle - \langle T(x), T(y) \rangle - \langle T(y), T(x) \rangle + \langle T(y), T(y) \rangle = \langle x, x \rangle - \langle x, y \rangle - \langle y, x \rangle + \langle y, y \rangle$$

As $\langle T(x), T(x) \rangle = \|T(x)\|^2 = \|x\|^2 = \langle x, x \rangle$ and similarly for y , by orthogonality $\langle x, y \rangle = \langle y, x \rangle = 0$, we conclude $\langle T(x), T(y) \rangle = 0$

Solution 7.3. (a) Either by solving the linear equations or finding the reduced row echelon

for the matrix, we may find a basis for this null space is $\left\{ \begin{pmatrix} 2 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 8 \\ 0 \\ -1 \\ 11 \end{pmatrix} \right\}$.

Now, using the Gram-Schmidt process, or simply by observation, we may find that

$$\begin{pmatrix} 8 \\ 0 \\ -1 \\ 11 \end{pmatrix} - \frac{16}{5} \begin{pmatrix} 2 \\ 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} \frac{8}{5} \\ -\frac{16}{5} \\ -1 \\ 11 \end{pmatrix} \text{ is orthogonal to } \begin{pmatrix} 2 \\ 1 \\ 0 \\ 0 \end{pmatrix}.$$

Now we normalize them and get $\frac{(2,1,0,0)^T}{\|(2,1,0,0)^T\|} = \frac{\sqrt{5}}{5}(2, 1, 0, 0)^T$, and $\frac{(\frac{8}{5}, -\frac{16}{5}, -1, 11)^T}{\|(\frac{8}{5}, -\frac{16}{5}, -1, 11)^T\|} = \frac{5}{\sqrt{3370}}(8, 16, -1, 11)^T$.

(b) First, we project on the orthogonal basis.

$$\frac{1}{5} \left\langle \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 2 \\ 1 \\ 0 \\ 0 \end{pmatrix} \right\rangle \begin{pmatrix} 2 \\ 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} \frac{6}{5} \\ \frac{3}{5} \\ 0 \\ 0 \end{pmatrix}, \text{ and } \frac{5}{674} \left\langle \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} \frac{8}{5} \\ -\frac{16}{5} \\ -1 \\ 11 \end{pmatrix} \right\rangle \begin{pmatrix} \frac{8}{5} \\ -\frac{16}{5} \\ -1 \\ 11 \end{pmatrix} = \left(\frac{168}{1685}, -\frac{336}{1685}, -\frac{21}{337}, \frac{231}{337} \right)^T$$

Then we sum them to get $\left(\frac{438}{337}, -\frac{135}{337}, -\frac{21}{337}, \frac{231}{337} \right)^T$

Solution 7.4. Clearly 1 remains as the first vector.

$$u_2 = x - \frac{\int_0^1 x \cdot 1 dx}{\int_0^1 1 dx} \cdot 1 = x - \frac{1}{2}.$$

$$u_3 = x^2 - \frac{\int_0^1 x^2 \cdot 1 dx}{\int_0^1 1 dx} \cdot 1 - \frac{\int_0^1 x^2(x - \frac{1}{2}) dx}{\int_0^1 (x - \frac{1}{2})^2 dx} (x - \frac{1}{2}) = x^2 - x + \frac{1}{6}.$$

$$u_4 = x^3 - \frac{\int_0^1 x^3 \cdot 1 dx}{\int_0^1 1 dx} \cdot 1 - \frac{\int_0^1 x^3(x - \frac{1}{2}) dx}{\int_0^1 (x - \frac{1}{2})^2 dx} (x - \frac{1}{2}) - \frac{\int_0^1 x^3(x^2 - x + \frac{1}{6}) dx}{\int_0^1 (x^2 - x + \frac{1}{6})^2 dx} (x^2 - x + \frac{1}{6}) = x^3 - \frac{3}{2}x^2 + \frac{3}{5}x - \frac{1}{20}.$$

Further normalizing them, an orthonormal basis is:

$$\left\{ 1, 2\sqrt{3}\left(x - \frac{1}{2}\right), 6\sqrt{5}\left(x^2 - x + \frac{1}{6}\right), 20\sqrt{7}\left(x^3 - \frac{3}{2}x^2 + \frac{3}{5}x - \frac{1}{20}\right) \right\}$$

8 Review problems

Remarks: The review problems are based on the questions from the previous finals. The length and format should be similar to the final, but it is not guaranteed. You can spare two and a half hours for the following question, or just take a glance and do the worthy ones, as you may already be familiar with some of them.

I sincerely wish good grades for all of you.

Problem 8.1. Determine whether the following statements are true or false. Explanation is **NOT** required.

1. A system of 3 linear equations with 4 unknowns must have a solution.
2. There exists some $A \in \mathcal{M}_{2 \times 2}(\mathbb{R})$ such that A, A^2, I_2 are linearly independent.
3. Every real symmetric matrix is diagonalizable.
4. Let T be a linear transformation from a finite-dimensional vector space to itself, and β_1, β_2 be two ordered bases for the space. Then $[T]_{\beta_1}$ and $[T]_{\beta_2}$ has the same reduced row echelon form.
5. If a matrix A is diagonalizable, so is A^T .
6. If n is odd, then $A \in \mathcal{M}_{n \times n}(\mathbb{R})$ has at least one eigenvalue.
7. If A is upper-triangular, then A^2 cannot be symmetric.
8. If $A, B \in \mathcal{M}_{n \times n}(\mathbb{R})$ and A is invertible, then AB and BA are similar.
9. Given U, V are two 4-dimensional subspaces of $\mathbb{R}^8$, there exist some bases of U, V respectively such that the union of the two is a basis for $\mathbb{R}^8$.
10. In a finite-dimensional inner product space V , if $\langle u, v \rangle = \langle u, w \rangle, \forall u \in V$, then $v = w$.

Problem 8.2. Solve the following linear system using Gaussian elimination.

$$\begin{pmatrix} 1 & 2 & 2 & 2 \\ 2 & 5 & 4 & 5 \\ 1 & 3 & 2 & 3 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 3 \\ 7 \\ 4 \end{pmatrix}$$

Problem 8.3. Let $V = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 \mid 2x + y - z = 0 \right\}$ be a subset of $\mathbb{R}^3$.

- (a) Prove that V is a subspace of $\mathbb{R}^3$.
- (b) Find an orthonormal basis for V and find the dimension of it.

Problem 8.4. Let T_m denote the linear transformation of projection onto the line $y = mx$ in $\mathbb{R}^2$.

- (a) Find the corresponding matrix of T_1 with respect to the standard basis.
- (b) Find the matrix representation of T_2 with respect to the ordered basis: $\left\{ \begin{pmatrix} 1 \\ 2 \end{pmatrix} \begin{pmatrix} 3 \\ 4 \end{pmatrix} \right\}$
- (c) Prove that for all m :
 - (i) The matrix representation of T_m with respect to any ordered basis is non-invertible.
 - (ii) There exist some linearly independent subset $S \subset \mathbb{R}^2$ for which $T_m(S)$ is linearly independent.
 - (iii) A_0 is similar to A_m , where A_m is the matrix representation with respect to the standard basis of $\mathbb{R}^2$.

Problem 8.5. Suppose sequences $\{x_t\}, \{y_t\}, \{z_t\}$ follows

$$x_{t+1} = 5x_t + 4z_t$$

$$y_{t+1} = 5y_t + 4z_t$$

$$z_{t+1} = 4x_t + 4y_t + 9z_t$$

for all $t \geq 0$ and $\begin{pmatrix} x_0 \\ y_0 \\ z_0 \end{pmatrix} = \begin{pmatrix} 12 \\ 6 \\ 6 \end{pmatrix}$.

- (a) Write down a matrix A such that $\begin{pmatrix} x_{t+1} \\ y_{t+1} \\ z_{t+1} \end{pmatrix} = A \begin{pmatrix} x_t \\ y_t \\ z_t \end{pmatrix}$.
- (b) Show that 1 is an eigenvalue of A , and diagonalize A .
- (c) Find $\{x_t\}, \{y_t\}, \{z_t\}$ in terms of t only.

Problem 8.6. A square matrix is said to be **nilpotent** if $A^k = 0$ for some positive integer k . The smallest such k is called the degree of A .

- (a) Show that $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ is nilpotent and determine its degree.
- (b) Find the determinant of a nilpotent matrix.
- (c) Show that if A is nilptent, then $I + A$ is invertible.
- (d) Let A be a $n \times n$ nilpotent matrix with degree k .
 - (i) Show that there exists $x \in \mathbb{R}^n$ such that $A^{k-1}x \neq 0$.
 - (ii) For such x , show that the set $\{x, Ax, \dots, A^{k-1}x\}$ is linearly independent.

Problem 8.7. Let $A \in \mathcal{M}_{n \times n}(\mathbb{R})$ such that $A^2 = A$. Prove that A is diagonalizable.
Hint: Consider possible eigenvalues of A , and the subspaces $\text{col}(A - I_n)$ and $\text{col}(A)$.

Solutions

Solution 8.1. (Explanation is NOT required.)

1. False: Any system may not have a solution, as it may contain equations like $0x = 1$.
2. False: We consider $A^2 - \text{tr}(A)A + \det(A)I_2 = 0$. Then they are not linearly independent.
3. True: This statement is difficult to prove. For formatting reasons, the idea is put at the end of the solutions.
4. False: Consider the projection $\mathbb{R}^2 \mapsto \mathbb{R}$, then changing the order of the basis will change the RREF.
5. True: If $A = P^{-1}DP$, then $A^T = P^T D^T (P^{-1})^T = P^T D (P^T)^{-1}$.
6. True: Consider the characteristic polynomial, which is of an odd degree. Then it must have at least one root in $\mathbb{R}$. As you can decompose it to linear and quadratic terms.
7. False: A can itself be diagonal.
8. True: $AB = A(BA)A^{-1}$.
9. False: This is true only when $U \cap V = \{0\}$.
10. True: Consider $u = v - w$, then $\langle u, v \rangle - \langle u, w \rangle = \langle v - w, v - w \rangle = \|v - w\|^2 = 0$. This gives $v = w$.

Solution 8.2.
$$\begin{pmatrix} 1 & 2 & 2 & 2 & 3 \\ 2 & 5 & 4 & 5 & 7 \\ 1 & 3 & 2 & 3 & 4 \end{pmatrix} \Rightarrow \begin{pmatrix} 1 & 2 & 2 & 2 & 3 \\ 0 & 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 \end{pmatrix} \Rightarrow \begin{pmatrix} 1 & 2 & 2 & 2 & 3 \\ 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \Rightarrow \begin{pmatrix} 1 & 0 & 2 & 0 & 1 \\ 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

Then, the solutions are $\{(x_1, x_2, x_3, x_4)^T \in \mathbb{R}^4 \mid x_1 + 2x_3 = 1, x_2 + x_4 = 1\}$, or equivalently,

$$\left\{ \begin{pmatrix} 1 \\ 1 \\ 0 \\ 0 \end{pmatrix} + c_1 \begin{pmatrix} -2 \\ 0 \\ 1 \\ 0 \end{pmatrix} + c_2 \begin{pmatrix} 0 \\ 1 \\ 0 \\ -1 \end{pmatrix} \right\}$$

Solution 8.3. (a) To prove that V is a subspace, the inclusion is clear, it remains to show that it is a vector space. That is the following three conditions:

- (1) 0 is in V , as $2 \cdot 0 + 0 - 0 = 0$;
 - (2) It is closed under addition, as $2x_1 + y_1 - z_1 = 0$ and $2x_2 + y_2 - z_2 = 0$ gives $2(x_1 + x_2) + (y_1 + y_2) - (z_1 + z_2) = 0$;
 - (3) It is closed under scalar multiplication, as $2x + y - z = 0$ gives $2ax + ay - az = 0$.
- (b) Since there are two free variables, the dimension is 2. By letting y and z to be 0

respectively, we have a basis of it being $\left\{ \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}, \begin{pmatrix} 1 \\ -2 \\ 0 \end{pmatrix} \right\}$.

Now, using the Gram-Schmidt process or by observation, we have $\left\langle \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}, \begin{pmatrix} 1 \\ -2 \\ 0 \end{pmatrix} - \frac{1}{5} \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} \right\rangle =$

0. Further normalize it we have an orthonormal basis being $\left\{ \frac{1}{\sqrt{5}} \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}, \frac{\sqrt{130}}{26} \begin{pmatrix} \frac{4}{5} \\ -2 \\ -\frac{8}{5} \end{pmatrix} \right\}$

Solution 8.4. (a) As $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$ are both mapped to $\begin{pmatrix} \frac{1}{2} \\ \frac{1}{2} \end{pmatrix}$, the corresponding matrix is $\begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix}$.

(b) There are two approaches, the first one is to calculate $T_2 \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ and $T_2 \begin{pmatrix} 3 \\ 4 \end{pmatrix}$, and then convert it using this ordered basis. As $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$ is mapped to itself, which is $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ with respect to this ordered basis, and $\begin{pmatrix} 3 \\ 4 \end{pmatrix}$ being mapped to $\begin{pmatrix} \frac{11}{5} \\ \frac{22}{5} \end{pmatrix}$ is $\begin{pmatrix} \frac{11}{5} \\ 0 \end{pmatrix}$ with respect to it, the matrix is $\begin{pmatrix} 1 & \frac{11}{5} \\ 0 & 0 \end{pmatrix}$.

The alternative approach is as follows. We let α denote the standard basis and β denote the new ordered basis. With respect to α , as $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$ are mapped to $\begin{pmatrix} \frac{1}{5} \\ \frac{2}{5} \end{pmatrix}$ and $\begin{pmatrix} \frac{2}{5} \\ \frac{4}{5} \end{pmatrix}$ respectively, $[T_2]_\alpha = \begin{pmatrix} \frac{1}{5} & \frac{2}{5} \\ \frac{2}{5} & \frac{4}{5} \end{pmatrix}$. Now, calculate $[Id]_\alpha^\beta [T_2]_\alpha [Id]_\beta^\alpha$ to get the same answer as above.

(c)

(i) Fix any m , as the line $y = -\frac{1}{m}x$ is mapped to the origin, which gives $null(T_m) \neq \{0\}$. Then, T_m is not injective, and hence not invertible.

(ii) We consider the set S with only one element v . It is enough to show that v is not mapped to 0 by any T_m . A simple example is to take any $(x, 0)$ with $x \neq 0$. i.e. $S = \{(1, 0)\}$, for example.

(If the question is to ask for a linearly dependent $T_m(S)$, then any linearly independent S with 2 elements will work, as $R(T_m)$ always has dimension 1.)

(iii) Notice that if we "rotate" the plane by some angle, then $y = mx$ will be the same as the original $y = 0$. Then, we consider a new basis of $\beta' = \left\{ \begin{pmatrix} 1 \\ m \end{pmatrix}, \begin{pmatrix} -m \\ 1 \end{pmatrix} \right\}$. T_m with respect to this basis is exactly the same as A_0 , as it sends the first vector (of the basis) to itself and the second vector to 0. Then, $A_m = [Id]_{\beta'}^\alpha A_0 [Id]_\alpha^{\beta'}$, which proves similarity.

Solution 8.5. (a) $\begin{pmatrix} 5 & 0 & 4 \\ 0 & 5 & 4 \\ 4 & 4 & 9 \end{pmatrix}$.

(b) Either by showing $\det(A - I) = 0$ or considering the characteristic polynomial, which is $\det(A - \lambda I) = -\lambda^3 + 19\lambda^2 - 83\lambda + 65 = (\lambda - 1)(\lambda - 5)(\lambda - 13) = 0$, we have 1 as an eigenvalue. (This factorization should not be difficult, as you are given that 1 is a root.)

The corresponding eigenvectors are $(-1 \ -1 \ 1)^T$, $(-1 \ 1 \ 0)^T$, $(1 \ 1 \ 2)^T$ respectively.

Then, let $Q = \begin{pmatrix} -1 & -1 & 1 \\ -1 & 1 & 1 \\ 1 & 0 & 2 \end{pmatrix}$, $Q^{-1}AQ = D = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 13 \end{pmatrix}$.

(c) Let $v_0 = \begin{pmatrix} 12 \\ 6 \\ 6 \end{pmatrix}$, then $v_0 = -4 \begin{pmatrix} -1 \\ -1 \\ 1 \end{pmatrix} - 3 \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} + 5 \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$.

Then, $\begin{pmatrix} x_t \\ y_t \\ z_t \end{pmatrix} = A^t v_0 = -4A^t \begin{pmatrix} -1 \\ -1 \\ 1 \end{pmatrix} - 3A^t \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} + 5A^t \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$
 $= -4 \begin{pmatrix} -1 \\ -1 \\ 1 \end{pmatrix} - 3 * 5^t \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} + 5 * 13^t \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 4 + 3 * 5^t + 5 * 13^t \\ 4 - 3 * 5^t + 5 * 13^t \\ 10 * 13^t - 4 \end{pmatrix}$ Which gives the explicit formulas of them.

Solution 8.6. (a) Clearly $A^2 = O$, $A \neq O$. Then it's nilpotent and the degree is 2.

(b) The determinant is 0. As $\det(A^k) = \det(A)^k = 0$ gives $\det(A) = 0$.

(c) We consider the equation $(A + I)(I - A + A^2 + \dots + (-1)^{k-1} A^{k-1}) = I + (-1)^{k-1} A^k = I$. This shows that $(I - A + A^2 + \dots + (-1)^{k-1} A^{k-1})$ is the inverse of $A + I$.

(d)

(i) This is clear, otherwise $(k - 1)$ will be the degree of A .

(ii) Assume that $\sum_{i=0}^{k-1} c_i A^i x = 0$. Apply A^{k-1} to both side, and note that all $A^{k+n} x$ for some non-negative n is zero, we have $c_0 A^{k-1} x = 0$, which means $c_0 = 0$. Similarly, by applying $T^{k-2}, \dots, T$ to both sides we have all c_i s are zero. Then they are linearly independent.

Solution 8.7. Consider an eigenvalue λ of A (assume it has one), then $A^2 v = \lambda^2 v = Av = \lambda v$ gives λ being 0 or 1. This gives us an intuition that A is equal to the identity outside of its null space.

We prove that $N(A) \cap R(A) = \{0\}$. This is because for any v inside the both the range and the null, $v = Ax = A^2 x = Av = 0$.

Let the whole space be V . Since $\dim(N(A) + R(A)) = \dim N(A) + \dim R(A) = \dim(V)$, and $(N(A) + R(A)) \subset V$. This suggests that $N(A) + R(A) = V$. That is, any $v \in V = u + w$, for some $u \in N(A)$ and $w \in R(A)$.

Now, inside $R(A)$ we have that $w = Ax = A^2 x = Aw$, which means A is indeed the identity. Clearly inside $N(A)$ we have $A = O$. Then, if we form a basis of V using a basis of $R(A)$

union a basis of $N(A)$, A can be represented by the matrix $\text{diag}(\overbrace{1, 1, \dots, 1}^{\dim(R(A))}, \overbrace{0, \dots, 0}^{\dim(N(A))})$. As the first $(\dim(R(A)))$ vectors are mapped to themselves, while the rest are mapped to 0.

Since for any basis, the matrix of the change of basis is invertible, A with respect to that basis is similar to our matrix above, which means A is diagonalizable.

Proof for 1.3:

This statement is the spectral theorem for real (finite dimensional) spaces, which says that any self-adjoint operator is diagonalizable. In real spaces, we have in general $\langle Ax, y \rangle = \langle x, A^T y \rangle$. This can be verified entrywisely, as they are equal to $\sum_{i=1}^n (\sum_{j=1}^n a_{i,j} x_j) y_i$ and $\sum_{j=1}^n (\sum_{i=1}^n a_{j,i} y_i) x_j$ respectively. After proper rearrangement, they are exactly the same.

Suppose the field is complex. For any eigenvalue λ (possibly complex). Note $Ax = \lambda x \Rightarrow \bar{x}^T Ax = \bar{x}^T \lambda x = \lambda$. Then, $\bar{\lambda} = \overline{(\bar{x}^T Ax)} = \bar{x}^T \bar{A}^T \overline{(\bar{x}^T)^T} = \bar{x}^T Ax = \lambda$, as $\bar{A} = A$ is a direct result from A is real. This shows that all eigenvalues of A are real.

Now, consider the character polynomial of A over $\mathbb{C}$. It has n (possibly with repetition) roots, which are the eigenvalues of A . From the previous discussion, they must be all real, thus A over $\mathbb{R}$ has at least one eigenvalue. We let it be v and thus define $\text{span}(v)$. Let $W = \text{span}(v)^\perp$, for any $w \in W$, $0 = \lambda \langle v, w \rangle = \langle Av, w \rangle = \langle v, Aw \rangle$. This show that $Aw \in W$, W is invariant under A . Thus, we may consider the linear transformation restricted to W , as it's real symmetric, we may repeat the previous discussion, and produce a new v' from W . Since the space is finite dimensional, by repetition finitely many times we may obtain an eigenbasis. Thus A is diagonalizable.